Exp. 4 Setting Up a Local Area Network

Date: 5/8/24

Aim:

To design, set up and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices verify connectivity using the ping command and demonstrate resource/data sharing accross the network

Equipment Required

No.	Equipment	Quantity
1	Personal Computers (PC)	4
2	Network switch	1
3	Ethernet cables (Cat5e/Cat6)	4

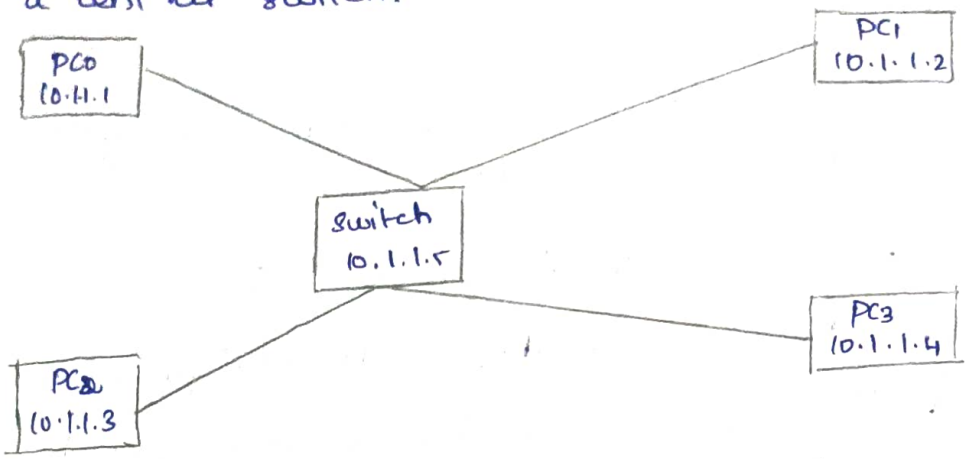
Theory

A LAN is a network that connects component and devices within a limited geographical area such as a home, office or laboratory, allowing them to share resources such as files printers and internet connections.

A star topology is used in this experiment where all end devices (PCs) are individually connected to a central network switch. The switch operates at layer 2 of the OSI model and forwards data frames between connected devices on their MAC address.

Network Diagram.

The LAN was implemented using a star topology as shown below with 4 PCs connected individually to a central switch.



IP Address Configuration Table

Device	IP Address	Subnet Mask	Connected to
PC_0	10.1.1.1	255.0.0.0	Switch Port Fo 0/
PC_1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC_2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC_3	10.1.1.4	255.0.0.0	Switch Port Fa 0/
switch	10.1.1.5	255.0.0.0	~~Swi~~ VLAN 1 Interfac

Implementation in Cisco Packet Tracer.

Since physical hardware was limited the LAN was also implemented and verified using Cisco Packet Tracer simulation software following the steps

Step 1: Place the Devices

1. Open Cisco Packet Tracer.
2. From the device category panel, select End Devices and drag 4 PCs onto the workspace.
3. Select Switches and drag 1 switch (e.g. 2960 series) on to the workspace.

Step 2: Cable the Devices

4. Click Connections → select Copper Straight-Through
5. Click each PC, then click the switch and select an available FastEthernet port to complete each connection.
6. Repeat for all four PCs. Green link lights confirm a successful physical connection

Step 3: Assign Ip Addresses to PCs

7. Click a PC → Desktop tab - Ip Configuration
8. Enter the static IP Address and Subnet Mask as per the IP table for each PC

Step 4: Configure the switch Management IP

9. Click the switch → CLI tab
10. Enter the following commands to assign a Management IP

```
enable
Configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown.
exit
```

Step 5: Verify Connectivity Using Ping.

11. Click a PC → Desktop tab → Command prompt
12. Use the ping command to test connectivity to the other PC's e.g. ping 10.1.1.2
13. Successful "Reply from" Messages confirm connectivity.

Step 6: Demonstrate Data Sharing (Server Simulation)

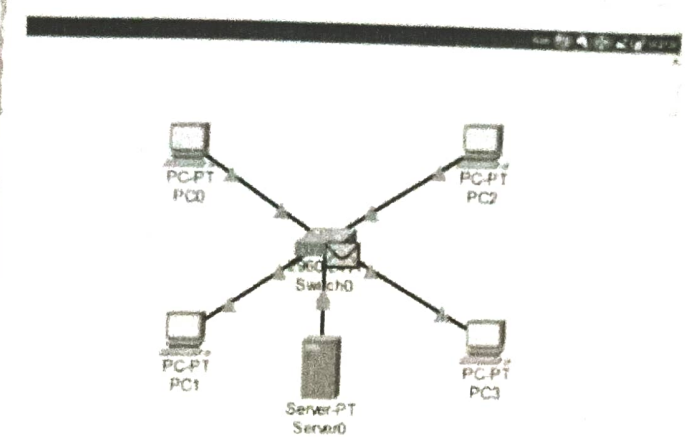
14. Add a server device (End devices → Server-PT) and connected it to the switch
15. Assign it an IP address via its Desktop → IP Configuration tab.
16. Open the server's Services (Config) tab and enable the HTTP service.
17. From another PC, open the web browser and enter the server's IP address to load the page confirming successful data transfer across the LAN.

Steps 7: Observe Packet flow in simulation Module

18. Switch to Simulation mode (bottom-right of the workspace).
19. Send a ping or web request between two devices and use the Play/Capture controls to step through the packet as it travel hop by hop through the switch, observing MAC address-based forwarding

7: Observation.

All four PCs were successfully connected to the switch in a star topology



Ack

Result :

Thus, setting up a Local Area Network is implemented successfully.